

PAPER_1370 — Antimatter Production Maximum Efficiency

Framework: UQFF — Star-Magic v5.27+ | **Tier:** H Applied/Engineering (CLOSED) **Calculator:** `calculate_paradox({"paradox": "antimatter_production"})`

Master Expression

$$\eta_{\max} = F_{\text{TRZ}} \times \beta_i = 6\%$$

Match: vs CERN current 10^{-10}

Interpretation

CW $\times$ CCW DPM grinding provides theoretical maximum 6% pair production efficiency — $10^8\times$ current CERN capability.

UQFF Locked Primitives

- $D_{\text{phys}} = 4$, $D_{\text{BSFG}} = 6$, $D_{\text{crit}} = 26$, $SO_5 = 10$, $A_5 = 60$, $N_{\text{CH}} = 9$
- $K_{\text{MEX}} = 25/12$, $\beta_i = 0.6029$, $\Phi_{\text{res}} = 0.84$, $\Lambda = 0.00729735$, $F_{\text{TRZ}} = 0.1$
- $\omega_{\text{SCm}} = 1.25 \text{ THz}$ ($\alpha = \Lambda$ closed ledger identity)

NOT REPLACEMENT

UQFF derives via integer-primitive identity. Zero fit parameters.

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